

Infosys Springboard Internship — Individual Contribution Documentation

Milestone 3: Week 5 & 6 — Treatment Effectiveness Analysis & Healthcare Analytics

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1. Executive Summary & Overview

During Milestone 3 (Week 5 & 6), the core objective focused on engineering treatment effectiveness analysis systems, medication outcome evaluation workflows, executive healthcare performance dashboards, and longitudinal readmission trend visualizers. St. Jude Medical Center's HealthForecast AI platform was enhanced with clinical outcome analytics, recovery curve tracking, ward occupancy indexes, and population health trend visualizers.

2. Key Modules & Individual Contribution

As an individual contributor, I designed, implemented, and verified the following technical modules:

- **Treatment Effectiveness Evaluation Hub:** Engineered clinician UI (/doctor/treatment-effectiveness) displaying treatment response categorization (72% Effective, 21% Partial, 7% Modifying) with active patient triage.
- **Medication Outcome & Efficacy Analysis:** Built analytical workflows evaluating patient response to insulin, metformin, and Lisinopril regimens with glycemic control and HbA1c trajectory visualizers.
- **Patient Recovery Telemetry Indexing:** Integrated dynamic recovery progress indexing (0-100%) and vital sign trend trackers within patient worksheets (/doctor/patients/:id).
- **Healthcare Performance & Outcome Analytics:** Built executive dashboard (/hospital-admin/analytics) tracking 30-day readmission benchmarks (14.2% vs 12.0% target), average length of stay, and recovery curve velocity.
- **Departmental Operations Visualizer:** Implemented ward-by-ward bed occupancy index (81.3% occupancy) and departmental readmission variance visualizers (/hospital-admin/performance).
- **Longitudinal Healthcare Trend Visualizer:** Built population health trend visualizer (/researcher/readmission-trends) tracking multi-month age group cohorts and stay duration correlations.

■ Milestone 3 Outcomes Verification Matrix

Required Task / Deliverable	Target Outcome	Implementation Component	Status
Implement treatment evaluation workflows	Treatment outcome distribution (72% effective) & triage	/doctor/treatment-effectiveness	COMPLETED
Generate recovery & treatment reports	Client-side exportable recovery telemetry & analytics	/doctor/treatment-effectiveness	COMPLETED
Develop medication outcome modules	Glycemic control & dosage adjustment metrics	/doctor/treatment-effectiveness	COMPLETED
Build healthcare performance dashboards	Hospital outcome KPIs & diagnostic recovery curves	/hospital-admin/analytics	COMPLETED
Generate patient outcome analytics reports	Departmental recovery velocity & outcome distribution	/hospital-admin/analytics	COMPLETED
Develop healthcare trend monitoring tools	Longitudinal multi-cohort trend visualizers	/researcher/readmission-trends	COMPLETED